

A New Hybrid Adaptive Deep Learning-Based Framework for UAVs Faults and Attacks Detection

Fadhila Tlili , Samiha Ayed , and Lamia Chaari Fourati 

Abstract—A resilient and guaranteed Unmanned Aerial Vehicles (UAVs) security framework should be designed to be secure against different types of attacks and faults. Recent developments have seen a proliferation of methods for improving UAVs security. Although, many studies proposed different approaches using artificial intelligence to enhance their security. Unfortunately, no study yet worked on an examination of a hybrid framework on UAVs faults and attacks and different architectures. Hence, our article aims to provide a prior detection results by proposing an hybrid adaptive framework for faults and attacks detection for UAVs applied on centralized and decentralized architectures. Our framework is based on two entry flows for faults and attacks in order to learn high-level features automatically from data. We validated our framework using deep-learning architectures. Finally, the empirical results show that our framework reached over 85% and 96,7% in term of accuracy for UAVs faults and attacks, respectively.

Index Terms—Unmanned aerial vehicles, attacks detection, faults detection, artificial intelligence, deep learning, UAVs security.

I. INTRODUCTION

OWING to the advancement of technology, Unmanned Aerial Vehicles (UAVs) [1] are aiding humans in different disciplines. UAVs proliferation in a wide variety of areas has been promoted by their dynamic re-configurability, rapid response, and simple deployment. As described in [2], they are used for various missions and domains including assistance, surveillance, and transportation in a variety of industries, including agriculture, forestry, ecological sustainability, and security. In the research domain, they are commonly known as flying IoT devices [3]. Initially, UAVs were used separately to perform as mono-UAVs. Nevertheless, multiple coordinated UAVs or multi-UAVs are now commonly deployed to accomplish critical operations. The introduction of multi-UAVs into the civilian market increased the demand for their usability. Despite technological advancements, the existence of faults is inevitable. Faults are probable to happen due to the fact that UAVs comprise different subsystems such as sensors, actuators, and other

elements that are prone to failure. Furthermore, unexpected attacks can occur within their working environment. Based on that, UAVs automotive cybersecurity [4] is now considered a fundamental issue comprising faults and attacks that may arise within UAVs components. As a result, UAVs security frameworks have witnessed a great demand in order to cover the UAVs security gap. Thus, the need for the development of robust and efficient UAVs faults and attacks detection and aiding methodologies has been increased. Faults and attacks detection aims to identify abnormal behavior. Under attacks, UAVs could be in an insecure state and will threaten the operation's integrity. This issue has been increased in the case of distributed UAVs which are originally implemented in order to accomplish the missions rapidly and efficiently. Several researchers studied faults and attacks detection using UAVs sensor and actuator data in order to get more intuitive analysis of the dynamic state. Despite this benefit, sensors are often too sensitive within the operating environment. For this reason, others simply use the data extracted from the MAVLink protocol. It is the most common open-source communication protocol that is currently in use for UAVs. It covers sensor/actuator data as well as network communications. It is customized for UAVs in particular to accurately represent the UAV operating system. In another direction, using flight logs is still effective for other UAV approaches to provide precise security solutions. UAVs flight logs include big amounts of extracted data. Owing to the amount of the gathered data, researchers proposed Machine Learning (ML) or Deep Learning (DL) approaches to build and evaluate their detection frameworks for UAVs faults and attacks. However, we argue that the current detection approaches are inherently flawed approaches. They are limited only to one goal faults or attacks detection. Others worked on both of them, but they did not reach good results yet. Moreover, despite their considerable efforts to detect UAVs faults and attacks, There is no study has been conducted on detection methods for both distributed and decentralized UAV systems.

Manuscript received 22 June 2022; revised 25 August 2023; accepted 29 August 2023. Date of publication 1 September 2023; date of current version 13 December 2023. Recommended for acceptance by E. Damiani. (Corresponding author: Fadhila Tlili.)

Fadhila Tlili is with the National School of Electronics, Telecommunications of Sfax, Sfax 3027, Tunisia (e-mail: fadhila.tlili@utt.fr).

Samiha Ayed is with the LIST3N-ERA, University of Technology of Troyes, 10300 Troyes, France (e-mail: samiha.ayed@utt.fr).

Lamia Chaari Fourati is with the Digital Research Center of Sfax (CRNS), SM@RTS (Laboratory of Signals, systems, aRtificial Intelligence, neTworkS), Sfax University, Sfax 3029, Tunisia (e-mail: lamiachaari1@gmail.com).

Digital Object Identifier 10.1109/TSC.2023.3311045

A. Research Motivations

Generally, the attacker could target an UAV without the associated Ground Control Station (GCS) being recognized. Our goal is to strengthen the GCS to detect whether a connected drone device is under attack. The detection is based on the UAV regular data patterns. We aim to work on failure detection with the same concept. For that, we investigate potential DL techniques for UAVs failure and attacks detection and propose a

new secure hybrid framework. More specifically, our framework is adaptive to distributed and decentralized architectures. From a computational point of view, the advantages of DL make it possible to achieve accurate results. Therefore, we look into other DL techniques that are embedded in a secure framework architecture that could be used for fault detection (FD) and attack detection (AD). This paper provides the following key contributions:

- 1) We present and discuss UAVs systems architecture and security issues, including faults and attacks.
- 2) We propose an hybrid-AFFA framework for UAVs to detect faults and attacks.
- 3) We evaluate our approach for distributed and decentralized UAV systems.

B. Roadmap

The rest of the paper is organized as follows. Section II presents a general overview of the key points covered in this paper. Section III discusses related work dealing with the detection of UAV faults and attacks. Section IV introduces our proposed adaptive and hybrid framework scheme. Section V describes our system model. Furthermore, Section VI presents our simulation results and evaluation. Finally, Section VII concludes the paper and discusses some future challenges.

II. UAVS SYSTEM OVERVIEW

In this section, we provide background information on UAVs-based systems. We focus on their different architectures and communication systems. We start by systematically introducing the general architecture of UAVs. Hence, we highlight existing UAVs cooperation mechanisms through communication systems for both mono-UAVs and multi-UAVs. Afterward, we list their security issues concerning faults and attacks. In addition, we show how artificial intelligence (AI) [5] techniques can be used to solve the security problems that we consider.

A. General Architecture of UAVs

UAV systems are usually made up of an unmanned plane, a ground control system (GCS), and communication links (CLs), while the mission planner is the person in charge of the GCS. The modern era of UAVs comprises various additional subsystems such as sensors, actuators, motors, structures, etc. Generally, there are two types of sensors: internal and external. The internal sensors continuously monitor the internal state's behavior like minimize power consumption, ensure a safe charging of the internal batteries, and detect fault conditions with motors or other system components. Meanwhile, external sensors generate different types of data during the flight simulation, including temperature, pressure, position, speed, velocity, acceleration, altitude, and longitude. Proper performance of all the aforementioned sensors is essential for a proper flight. Usually, the flight begins by altering the power engines to provide the required translational motions. Therefore, UAVs will lift, hover, tilt left/right to give a (*roll*) value, forward/backward for (*pitch*) value, or turn in on themselves for a (*yaw*) value. Thus, the key components of the UAV must correlate and cooperate in order to

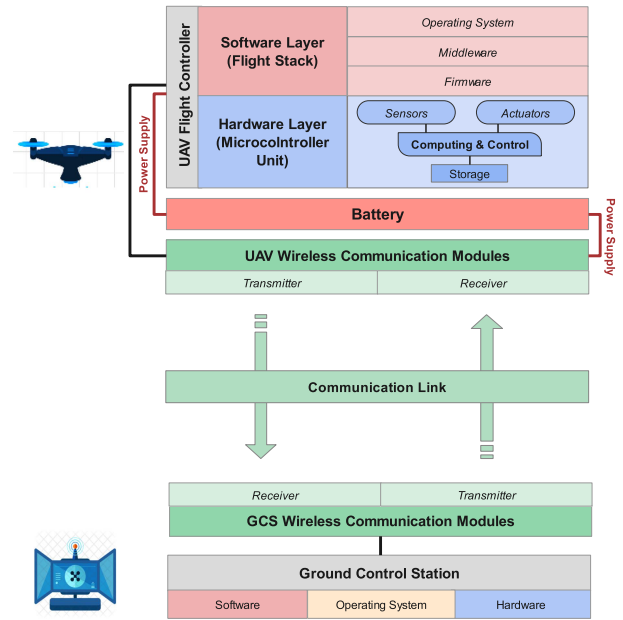


Fig. 1. The general architecture of an UAV.

maintain the desired state and ensure safe flight. In order to allow remote UAV flight, the GCS remotely controls and monitors them during their missions via wireless communication.

Hence, UAVs are able to integrate sensing, perceiving, assessing, communicating, scheduling, and decision-making. In addition, UAVs have the ability to respond to situations and fulfill their tasks as assigned via the communication system.

Fig. 1 depicts the architecture of an UAV that includes:

- **Hardware architecture [6]:** Among the things that make up the actual hardware of a UAV are a flight controller (FC), batteries that can be charged, actuators, wireless communication, and a set of sensors. These sensors include GPS, inertial measurement units (IMU), and accelerometers for things like orientation and location. Typically, sensors provide physical measurements of the surrounding environment, such as height, speed, and temperature. These measurements are turned into data by the flight controller and sent to the operator.
- **Software architecture [7]:** UAVs software design is layered on three primary layers: firmware, middleware, and operating system (OS). The integration of these layers constitutes the flight stack. The firmware is the lowest layer of the flight stack, and the OS is the highest layer, being a real time operating system (RTOS). Meanwhile, the middleware is responsible for guaranteeing appropriate flight control by regulating the operating system as a distributed embedded system.
- **Communication links [3]:** These links enable wireless communication between the GCS and the UAV. They ensure a safe, bidirectional transmission between the aircraft and the GCS during the flight operation. Communication channels are divided into two categories: data communication and control communication. Through data communication, the UAV transmits data signals to the GCS, such as telemetry and status data. While in control communication,

the GCS transmits instructions and control signals to the UAV.

Generally, UAVs operates in different platforms including low altitude platforms (LAPs) and high altitude platforms (HAPs). The LAPs operate in a low-altitude that typically do not exceeding several kilometers. Meanwhile, HAPs performs in a stratospheric altitude in typically quasi-stationary locations. Although HAP-based communications offer wider coverage and longer endurance. The diverse capabilities and placement options of UAVs offer a unique opportunity to establish multi-tier heterogeneous aerial networks.

B. UAVs Communication Networks

UAVs communication [8] is defined by the transmission between an UAV and another end node, which is referred to as UAV-2-X communication. To maximize their efficiency, it is necessary to implement appropriate communication networks using reliable protocols and security mechanisms. It is necessary to mention that the presence of an UAV network does not necessarily indicate the presence of a swarm. Based on this, mono-UAVs messages can be exchanged with the base stations, antennae, or any other equipment. Meanwhile, aerial communication networks differ from mobile and vehicular ad hoc networks in multi-UAVs. It is based on node mobility and the established topology. As a result, UAVs control systems can be centralized or decentralized. In centralized UAVs systems [9], every single UAV serves as a station. This architecture relies on motion capture systems and a central controller. As a result, it is suitable for simple and straightforward operations. However, every communication across UAV-2-UAV must pass through the GCS using this design. Meanwhile, decentralized UAVs systems [10] are represented by a group of flying multi-UAVs communicating as UAV-2-UAV without routing information to the GCS. This system relies on the internal logic of the backbone mono-UAV to coordinate movement and data transmission between the members of the group that constitutes multi-UAVs. In this case, only one backbone UAV exchanges data with the GCS. However, the centralized UAV routing leads to a delay in data transfer. As a result, it is not adequate for long-distance communications. Meanwhile, a decentralized architecture is more efficient owing to the established collaboration of multi-UAVs. Hence, we have to choose the architecture that will be used depending on the flight mission to meet the communication needs of autonomy and scalability. Hence, UAVs use outing protocols in order to guarantee a reliable data transmission between UAV nodes. There are two well known routing protocols that are commonly within the UAV environments:

- *MAVLink protocol* [11] [12]: is defined as a lightweight point-to-point communication protocol. It is mainly employed to exchange control and telemetry data in UAV-2-GCS operations. For real-time applications, the MAVLink protocol employs bidirectional communication via wireless channels.
- *UranusLink protocol* [13]: is a packet-oriented protocol deployed for UAV-2-GCS wireless communications. The design of this protocol fulfills radio communication

requirements, including data throughput and low latency, which makes it suitable for aerospace applications. However, uranusLink is appropriate for particular UAVs with a low flying altitude.

C. UAVs Security Issues

Despite their benefits, UAVs suffer from many vulnerabilities. To deal with UAV security, researchers examined several UAV attacks and faults that led to severe consequences. The most frequent attacks are like jamming attacks, spoofing attacks, and distributed denial of service attacks (DDoS) [14]. The jamming could include GPS jamming, control stream jamming, and data stream jamming. For spoofing it is equivalent to a jamming but more sophisticated. The DoS attacks are common direct availability attack implemented during the UAV mission. These attacks sneak through different means using planned entry points and cause different damages. However, these entries are the same components that could face different types of faults. They cause inter-security issues such as engine failure, aileron(s) failure, elevator failure, and rudder failure. The moment the failure occurs, the UAV error status will be on, and the UAV will become disabled. Thus, security frameworks should take into account hybrid possibilities of attacks and faults on UAVs. The interconnection between privacy and public safety in the context of UAVs has been seen as a prominent and ongoing concern by researchers in [15]. The ongoing effort of regulating the utilization of aerial platforms is currently under progress because of UAVs security and privacy issues. The inconsistent and unauthorized execution of UAVs attacks can result in significant risks to public safety. Still, UAV-assisted wireless communications encounter security and privacy threats due to the untrusted broadcast features and wireless transmission of UAV networks. These two issues pose a significant threat to system security and privacy. There are three types of attackers or adversaries commonly arise. These issues are assessed and described in [16] including malicious spectrum provider, malicious spectrum buyer, and malicious trusted third party.

D. AI Techniques for UAVs Faults and Attacks Detection

Most research studies supported AI in UAVs detection of faults and attacks. They shared the goal of reproducing an approach using AI tools and algorithms based on ML or DL methods. These AI algorithms are developed to build a theoretical and experimental proof with countermeasures to enhance UAVs security. The proposed detection techniques are classified as statistical-based, knowledge-based, or machine learning-based. Due to learning and training units, ML algorithms surpassed non-ML alternatives. However, [30] found that ML algorithms' performance is not sufficient to provide a suitable solution for FD and AD. Based on this, additional sophisticated DL algorithms were proposed to directly address UAVs fault and attack problems as a whole, like the one proposed in our paper. DL techniques have proven to be a reliable tool that overcomes ML limits, especially in UAVs fault and attack detection frameworks. Due to their deep structure, DL approaches are great at automatically learning complex features from raw data without

relying on feature engineering. Moreover, these approaches are intended for extracting valuable information from large amounts of data.

III. RELATED WORKS

Regardless that the research issue of UAVs faults and attacks detection is still quite new, there are some initiatives and proposed frameworks. Certain detection frameworks mainly focused on specific areas, such networks attacks detection or system failure detection, while others take a more balanced view to the security of the aircraft's system from both faults and attacks. Recent research in [11] proposed a novel solution for multi-UAVs networks applied on decentralized architectures. The developed system aims to secure UAVs and identify attacks using data packets from MAVLink communication protocol. The based model received four sequential data packets that consisted of attack free, heartbeat flood, ping flooding, and request flooding for software and hardware environments. This system model is set up on DL algorithm by implementing recurrent neural networks (RNN) [30] using long short term memory (LSTM) approach. It is commonly used in fault and attack detection models based on sequential data. Yet, with their successful detection results, the system has been validated only on injection attacks, which do not ensure its efficiency on other attacks. It considers only UAVs networks attacks. In addition, the authors assert their lack of sensors and actuators data, and they were not able to enhance the performance of their system. A recent work [21] has shown superior results via a proposed framework for UAVs network attacks detection based on DL techniques. The authors developed an hybrid technique based on LSTM-RNN models. The models are divided into two units, one located inside the flying Ad-Hoc networks (FANET) and the second unit is located in the GCS. The evaluation were done on a variety of datasets such as KDDCup 99 [31], [32], NSL-KDD [33], UNSW-NB15 [34], Kyoto [32], CICIDS2017 [35], and ToN_IoT [36]. The chosen datasets cover a wide range of attacks and validate the proposed approach. Each data-set has its own unique value and number of records. However, it is limited on attack detection, with a low accuracy on most attacks. An additional work in [17] supported the ongoing issue of UAVs attacks. The authors developed an intrusion detection system (IDS) for UAVs anomalies. The system has been validated on two different attacks extracted from RADAR records with normal or benign and abnormal sets of data. The system collects network data from sensors through distributed or centralized capturing devices based on the established topology. These data were processed to be fed to the neural network (NN) model. Researchers combined NN with LSTM to design an accurate detection model. LSTM-based detection is simple and deals easily with time series data. The results were prominent where the overall accuracy was high with different intensities that identify the attacks. Other researchers assumed that UAVs fault and attack detection problems could be solved using another RNN model, like the gated recurrent unit (GRU). GRU models are faster to learn and do not require memory units. Inspired by the above method, [26] built a detection scheme for UAVs abnormal behavior under attacks using

GRU that support good memory. This proposition is established in order to adapt to the characteristics of the angle data in UAVs. Based on this, the authors proposed to combine classic RNN and LSTM models in order to take in consideration the confusion. The proposed framework detects attacks using a slight offset of abnormal behavior to improve the accuracy. To initiate the process, three different neural networks including simple RNN, LSTM and GRU are used to learn and construct a trustworthy detection model. In addition, the authors used a specific estimation algorithm called direction of arrival (DOA) algorithm. It aims to gather larger numbers of training data to help in choosing a suitable threshold. In fact, this operation is the main limit of the proposed system because it is time-consuming. The algorithm takes a long time to determine the most suitable threshold value. The validation of this method indicates that the accuracy of the suggested detection framework is high only for UAVs spoofing attack detection. Authors in [22] raised an enhanced GRU method for UAVs fault detection based on residuals and threshold. The approach includes data pre-processing, estimation model, and detection model. The detection model is based on the residuals between the real and estimated data. The approach was good, but not sufficient to reach the desired goal of the authors. Likewise, [28] investigated the unsupervised mining methods based on k-means and LSTM. This method allows for accurate detection of time dependencies. The validation metrics were based on dynamic threshold distance calculations and tested using high-dimensional data. However, findings were insufficient even after employing LSTM in order to improve the detection sensitivity. Nonetheless, when the model was exposed to an unexpected attack, the reconstruction error increased at a rate of 20%. Meanwhile, only few studies addressed the hybrid detection of UAVs faults and attacks. Several other studies have been conducted for networks-based IDS for UAVs, including [18]. This study developed a Deep Learning (DL)-based IDS for networked UAV anomalies and illicit attack detection. The proposed scheme thwarts attacks on UAVs networks in both uni-directional and bidirectional flow-supervised learning modes. Other DL-based IDS solutions with enhanced generalization proposer ties have been proposed in [20]. They used a hybrid solution based on a group of convolution networks and snapshot ensemble learning. The IDS automatically learns data patterns in UAV communications and analyzes them. Hence, the core detection system has been evaluated using time-series data analysis to detect point abnormal behavior. Authors in [29] focused on UAVs-based IDS hardware attack detection based on malicious activities over sensors. They proposed a cyber-physical layer attack detection solution, especially on sensors based on the physical system data, and applied statistical analysis techniques. The authors considered spoofing, data injection and jamming attacks on the UAV sensors to validate their framework. They applied a residual attack detection mechanism based on a state estimator. Another study [27] conducted UAVs networks attacks detection via an adaptive, lightweight IDS to identify intruders and ensure the safe return of the drone. The authors analyzed the networks to examine network safety and identify the state of security in case of internal or external attacks. Then a DL-based model combined with Self-Taught Learning (STL) with

TABLE I
RELATED WORK COMPARATIVE TABLE

Ref - Year	System Architecture	Dataset	Technique	Efficiency	FD	AD	Adaptive
[17] - 2022	Distributed UAVS-IDS for network security	Private RADAR records	Wavelet Leader Multifractal, LSTM	Poor	X	✓	X
[18] - 2022	Network-based IDS for UAVs attacks detection	UAV-IDS-2020 dataset	CNN - ConvNet model	Good	X	✓	X
[19] - 2022	AI onboard IDS for UAV sensors attacks	KDD99 dataset	CNN	Good	X	✓	X
[20] - 2021	DL-based IDS for UAVs network and communication channels attacks	Simulated dataset	CNN	Good	X	✓	X
[11] - 2021	Decentralized MUVIDS framework for UAVs MAVLink attacks	MAVLINK Private Dataset	RNN-LSTM	Acceptable	X	✓	X
[21] - 2021	Hybrid technique for decentralized UAVs	KDDCup99, NSL-KDD, UNSW-NB15, Kyoto, CICIDS, ToN-IoT	LSTM	Good	X	✓	X
[22] - 2021	Centralized UAVs fault estimation	Not accessible	GRU	Poor	✓	X	X
[23] - 2021	Hybrid Framework for UAVs abnormal behavior detection	UA, ALFA	Autoencoder	Acceptable	✓	✓	X
[24] - 2020	Secure IDS for UAVs based attacks detection including failure detection	ALFA	KLD + ANN	Good	✓	✓	X
[25] - 2020	Novel IDS for UAVs Flight anomalies and failures	Simulated dataset	One-Class Support Vector Machine (OC-SVM)	Good	✓	X	X
[26] - 2019	Centralized UAVs abnormal behavior detection scheme	UA Dataset	RNN, LSTM, GRU	Good	✓	✓	X
[27] - 2019	Adaptive, lightweight IDS for UAVs Safety	Simulated Data	Self-Taught Learning + multi-class SVM	Acceptable	✓	X	✓
[28] - 2018	Centralized UAVs attack detection scheme	German dataset	LSTM	Acceptable	✓	✓	X
[29] - 2017	UAVs-based IDS for sensors attacks detection	Simulation dataset	Bayesian network	Poor	X	✓	X
Our Paper - 2023	AHFFA-UAVs	UA, ALFA, CAIDA	LSTM, BLSTM, GRU	Excellent	✓	✓	✓

multi-class SVM is used for known and unknown attack detection. Meanwhile, a self-healing IDS recovery phase using a deep-Q algorithm for dynamic path learning is launched to facilitate the drone's safe return home. Researchers in [24] investigated the security of UAVs system by introducing a hybrid IDS for UAVs attacks detection. They used the Kullback-Leibler divergence (KLD) approach and the neural networks (ANN) conventional model to detect UAVs attacks. The detection process is based on the external sensor data and the internal data transmitted through sensors to the ground station control. In [19], a new fully developed AI-based IDS that operates onboard the UAV has been proposed to detect and potentially mitigate attacks (e.g., GPS spoofing and jamming). In contrast to other proposed systems, utilizing an IDS agent on board facilitates the identification of attacks. In [25], an IDS that adopts novel detection techniques to identify security attacks was suggested. Unlike intrusion detection, novelty detection gives researchers the potential to learn from a training dataset without anomalies. This facilitates using existing flight records to evaluate what is typical for the UAV the IDS will behave on. In this context, [23] focuses on the existing faults and attacks approaches taking in consideration the aspects of UAVs. The authors suggested a model using a stacked unsupervised learning model [37]. It is based on a deep auto-encoder that achieves a low loss of reconstruction. The area under the curve (AUC) metric was used to validate this approach on two datasets containing diverse types of UAV faults and attacks. However, this method applies the model independently for faults and attacks, with a few modifications. Besides, the approach performance is low due to the high computational load. However, the complexity increases when UAVs are involved in more sensitive operations that require a higher level

of security. Meanwhile, most of previous studies have been limited to different stages being addressed simply for attacks or faults. Even for hybrid proposed frameworks, they perform the detection of faults and/or attacks using different executions. Additionally, the nature of UAVs networks requires the adoption of adaptable security processes, primarily for attacks detection. In summary, most existing adaptive IDS for UAVs are based on traditional supervised ML algorithm and supervised-based models. Therefore, it is necessary to design and develop a new DIDS to detect attacks in UAVs that consider all the current aspects and challenges. A resilient and adaptive framework should be designed to be secure and stable for UAVs. Based on this, the main objective of this study is to propose a new adaptive hybrid framework for both UAVs faults and attacks detection. The proposed solution overcomes the existing limits of past studies using DL-models. Table I provides a comparative summary of previous studies.

IV. PROPOSED HYBRID-AFFA FRAMEWORK FOR UAVS: SCHEME DESIGN

This section presents the scheme design of our proposed Hybrid Adaptive Framework for Faults and Attacks detection for UAVs named "hybrid-AFFA framework". The overall architecture is illustrated in Fig. 2 and Algorithm 1. Starting with preprocessing, the framework is composed of three main phases: monitoring, training based on DL-models, and detection. Our hybrid-AFFA framework is built to be adapted to both UAVs centralized and decentralized UAVs architectures. Since they have to communicate with the ground base station, a centralized architecture is based on mono-UAVs sending data directly to the

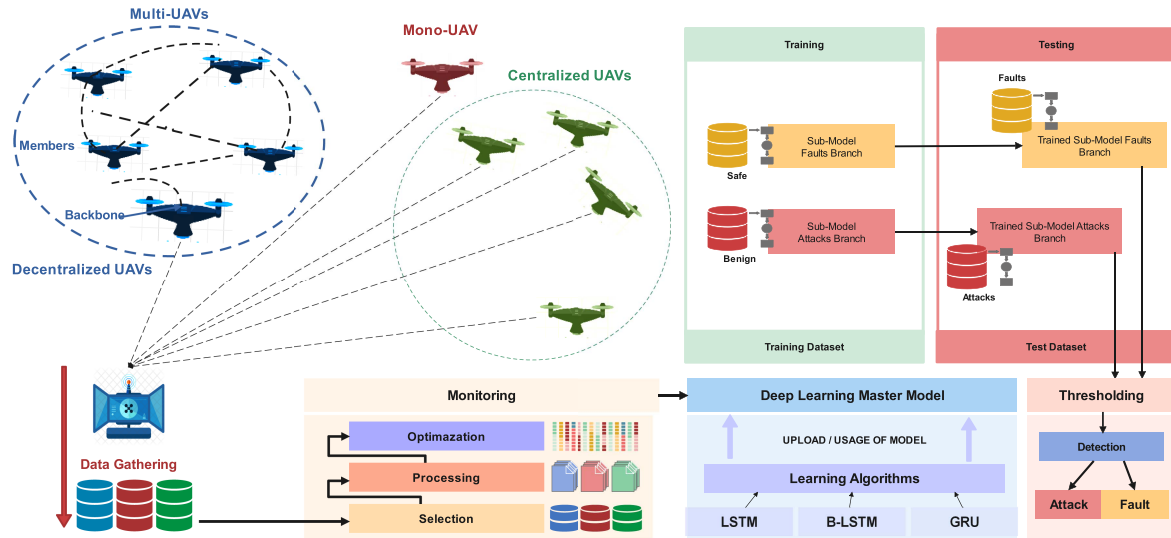


Fig. 2. The architecture of our proposed hybrid adaptive framework for faults and attacks for UAVs.

GCS through communication links. In the decentralized case, UAVs are set by different groups of Multi-UAVs with a backbone linked to the rest of the UAVs members with a bidirectional communication link with the GCS. In both cases, during the monitoring phase, we gather data from GCS to extract patterns from sensors, OS, communications links, and CS. Generally, our detection framework relies on the data that exist in all UAVs categories. For accurate detection, feature selection and prioritization of patterns are important. In addition, data processing is an essential part of generating the input features to avoid the curse of dimensionality and extract key features. After setting-up engineered data collection, the framework launches the second step of monitoring, which is optimization to solve the class imbalance problem and addressing the overfitting technique. Based on this, our framework will be adapted and open on different UAVs testing resources to enhance the framework's performance. Subsequently, we train our DL-models, which are designed in a novel scheme to detect both UAVs faults and attacks. The traditional detection models are built using different RNN approaches. They enable internal state adjustment in accordance with the output of the previous state, thus generating a feedback structure. Nevertheless, RNNs are prone to the gradient vanishing problem. To overcome this problem, we propose using LSTM, bidirectional-LSTM (B-LSTM), and GRU as they can retain memory for an extended duration. The DL-model is divided into a master model and two sub-models, inflow for faults and attacks. As mentioned previously, the DL-master is a shared-based model that yields three architectures of validation models. Meanwhile, the two sub-models entail the main structure of the master and apply it based on the type of features. We apply the training phase to these two sub-models based on normal, safe flight states without faults and benign flight states without attacks to generate an hybrid adaptive global model. As a result, the trained sub-models will be aggregated and the DL-master will be trained using: LSTM, B-LSTM, and GRU for centralized and decentralized different faults and attacks datasets. Consequently, the framework starts the detection phase

based on the thresholding function. We validate the efficiency of abnormal behavior identification and compute the detection performance. In the case of fake detection or alerts, our framework overcomes this issue through enhancing the thresholding. We implement a dynamic error and smoothing threshold combined with a function for mitigating fake thresholds.

V. AHFFA-UAVS SYSTEM MODELING

This section describes the overall process of our AHFA-UAVs framework in details. Our method is divided into several parts including system monitoring, DL-model, sub-models training and detection based on thresholding with decision-making. The overall pipeline of AHFFA-UAVs is shown in Fig. 3.

A. System Monitoring

This subsection presents the monitoring phase of our framework. We focus on dealing with common datasets of various faulty and anomalous circumstances. We won't delve into unrecognizable features that exist only in typical UAVs categories. These analyses are done to prepare the dataset for proper training of the DL-model. This phase includes three main steps responsible for preparing the input of data patterns, which are as follows:

1) *Data Gathering and Selection:* We believe that gathering data from flight logs that hold faults, communication issues, or have been attacked is extremely hard and takes a lot of resources. It's essential for this work to highlight that we have made all the flight test and validation data operational. Our framework is able to deal with different data types. As a result, data gathering is an initial step for validation, while the rest is fundamental to make the acquired data ready. In our case, the GCS data collector module receives all the flight logs for different systems architectures. The data is then stored in clusters to be prepared and passed through the pipeline of our framework. Meanwhile, data selection is essential to strip different pattern packets and extract the necessary features from data clusters. Extracting features aims to transform the input data into a vector of features

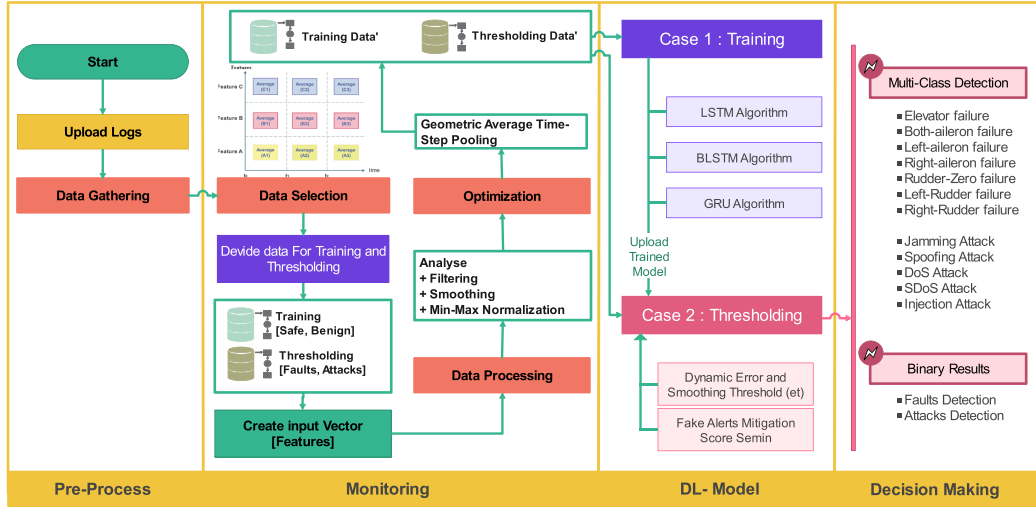


Fig. 3. Our adaptive hybrid UAVs framework for faults and attacks pipeline for the system operations.

that contains data patterns. Generally, feature selection aims to choose the best data pattern. To perform this step, we apply a selector module that performs based on criteria of selection and extrudes the rest. Our framework selects features based on hardware generality and sensor stability. The hardware generality seeks to exclude the unique features by applying the inclusion criterion. This aims to build an easy framework regardless of the hardware that exists only in particular UAVs. The inclusion criterion cluster works on the data that holds common characteristics extracted from the sensors [actuator control patterns, GPS location report patterns]. For the sensor's stability, they aim to exclude the constant features that do not provide any meaningful information. These features are frequently recorded in the case of lost or stable values that stay frozen for a long duration. By excluding these features, we decrease the curse of dimensionality and speed up the processing. After applying the selection procedure, we re-store the samples for processing and optimization.

2) *Data Processing:* Flight logs usually contain decent noise like vibration from motors and wind, which may influence the detection. This usually happens when data is extracted from sensors. We denoise these logs by applying several filtering functions to remove outliers, missing-values, and offsets. Among processing functions, there are functions incorporating fill-missing, fill-outliers, and smoothing to remove undesirable noise or high variance from data. Generally, we clean the data and rescale the features to get a promising indicator and enhance the framework performance to easily deploy the DL-model for the detection. Therefore, we apply a min-max normalization function to resize the data-frames. This function changes the range of values between [0] and [1], giving the design a single set of values. This approach preserves the range (minimum; maximum) and provides a sequential dispersion, such as the deviation and the variance.

3) *Data Optimization:* Data processing and training may be overlapped, that is why optimization is required. Typically, through optimization, we can ensure the efficiency of the performance. We optimize the duration of the features because

they are recorded at different times. Data optimization relies on timestamp geometric average-pooling. We contribute the key to this function to enhance the detection efficiency and consider the compounding that occurs from time to time. This function aims to transform the unequal values of features ($Scaled_F$) in a parallel timestamp into unified fixed sizes at a defined timestamp between $[t, t + 1]$. We calculate the geometric average of a given value in a given interval, as detailed in (1).

$$\text{Geometric_Av_T}(F_x) = \left(\prod_{x=1}^n F_x \right)^{\frac{1}{n}} = \sqrt[n]{F_1 F_2 \cdots F_n}. \quad (1)$$

In this function, $\prod$ denotes the geometric average of x th value of the feature F and $F(x)$ is the value of features-set that average from $x = 1, 2, \dots, n$.

B. DL-Model and Sub-Models Training

After reviewing a variety of papers on UAVs faults and attacks, we were inspired to build the DL-model based on diverse architectures. We have sufficient evidence that RNNs and their ensemble implementations (as described in Section IV) are the most suitable. The main idea is to use different types of RNNs, like LSTM, B-LSTM, and GRU, to try to get an accurate hybrid adaptive detection. This model is linked to two sub-models to perform training, detection, and validation. These models are able to learn from extensive data series and provide information for a long time. In addition, one of the major benefits of the B-LSTM is the relationship between future and previous learning units, which represents the bidirectional term. The LSTM is a sub-type of the RNN model that learns long-term relevant time-dependencies effectively, decreases gradient increase, and minimizes vanishing difficulties during learning. The LSTM comprises three main gates which are: the forget gate f_t , the input gate i_t , and the output gate o_t . There is an additional memory cell followed by an intermediate gate where t represents the referred timestamp. The three gates control the information flow in and out of the LSTM structures to manage and secure data. As

Algorithm 1: Hybrid-AFFA Framework for UAVs.

```

1: Initialization:
2: Initialize Framework
3:
4: Preprocessing Phase:
5:  $x_{\text{raw}} \leftarrow \text{GatherDataFromGCS}()$ 
6:  $p_{\text{patterns}} \leftarrow \text{ExtractPatterns}(x_{\text{raw}})$ 
7:  $f_{\text{features}} \leftarrow \text{DataProcessing}(p_{\text{patterns}})$ 
8:  $f_{\text{norm}} \leftarrow \text{NormalizeData}(f_{\text{features}})$ 
9:
10: DL-Model Training Phase:
11:  $M_{\text{master}} \leftarrow \text{CreateMasterModel}()$ 
12:  $F_{\text{sub}} \leftarrow \text{CreateSubModel}()$ 
13:  $A_{\text{sub}} \leftarrow \text{CreateSubModel}()$ 
14: for each epoch in TrainingEpochs do
15:    $x_{\text{normal}} \leftarrow \text{GetNormalData}()$ 
16:    $x_{\text{benign}} \leftarrow \text{GetBenignData}()$ 
17:    $\text{TrainModel}(F_{\text{sub}}, x_{\text{normal}})$ 
18:    $\text{TrainModel}(A_{\text{sub}}, x_{\text{benign}})$ 
19:  $M_{\text{global}} \leftarrow \text{CombineModels}(M_{\text{master}}, F_{\text{sub}}, A_{\text{sub}})$ 
20:
21: Detection Phase:
22:  $y_{\text{detection}} \leftarrow \text{DetectAnomalies}(M_{\text{global}})$ 
23: performance  $\leftarrow \text{EvaluateDetectionPerformance}(y_{\text{detection}})$ 
24:
25: Enhance Detection:
26: threshold_dynamic  $\leftarrow \text{CalculateThreshold}(\text{performance})$ 
27: threshold_smooth  $\leftarrow$ 
   SmoothThreshold(threshold_dynamic)
28: threshold_final  $\leftarrow$ 
   MitigateFalsePositives(threshold_smooth)
29:
30: Results and Evaluation:
31: PrintDetectionResults( $y_{\text{detection}}$ )
32: EvaluateEfficiency(performance)
33:
34: Threshold Calculation and Adjustment Functions
35: function CALCULATETHRESHOLD(performance):
36:   threshold  $\leftarrow$ 
     BaseThreshold + (performance  $\times$  ThresholdMultiplier)
37:   return threshold
38: function SMOOTHTHRESHOLD(threshold):
39:   smoothed_threshold  $\leftarrow \text{SmoothingFactor} \times \text{threshold} +$ 
     (1 - SmoothingFactor)  $\times$  prev_smoothed_threshold
40:   return smoothed_threshold
41: function MITIGATEFALSEPOSITIVES(threshold):
42:   mitigated_threshold  $\leftarrow \text{threshold} + \text{MitigationFactor}$ 
43:   return mitigated_threshold
44: END

```

a result, the LSTM design allows the system to save weighted data from the previous learned state. A cell in the LSTM is made up of two activation functions: the sigmoid (σ) and the tangent $\tanh$ functions. They are both combined to update the cell states. The output cell of the DL-model is connected to the framework output cell. However, the network will learn

safe and benign patterns in order to detect the deviations within the thresholding phase. Since it operates on raw data, LSTM as a DL algorithm makes regularization easier. Nevertheless, our framework also uses the B-LSTM commonly known as the bidirectional recurrent neural network [38] as an enhanced architecture of the LSTM. To overcome the limitations of LSTM conventional and sequential modeling, we rely on the B-LSTM. The conventional issue is defined by having a one-way network and displaying only the current state associated with the past context. Based on this, B-LSTM represents our second altered DL-model architecture, where the LSTM layers are replaced with the B-LSTM cell units. The B-LSTM layers bidirectionally learn the dependencies and movements between sequence patterns. This fact is essential for learning discriminative features at each time step. The B-LSTM generates two hidden states for each timestamp t as h_t^f to determine the forward layer hidden state at time step t and h_t^b to determine the backward layer hidden state at time step t . Therefore, the output layer is connected by merging the hidden units of the forward and backward layers. Yet, there is no connection between forward and backward cell states. Based on this, prospective data will be provided if no restrictions are anticipated. In general, a set of features X that corresponds to a target vector y are fed into the B-LSTM model. They produce an input gate vectors $[i \rightarrow j | i \leftarrow j]$, a forget gate vectors $[f \rightarrow j | f \leftarrow j]$, memory cell state vectors $[c \rightarrow j | c \leftarrow j]$, output gate vectors $[o \rightarrow j | o \leftarrow j]$, and hidden state vectors $[h \rightarrow j | h \leftarrow j]$. Therefore, the B-LSTM efficiently and precisely detects UAVs faults and attacks by looking at the long changes in flight. In addition to the BLSTM, we added the GRU as a third altered architecture to the DL-model. The GRU is a specialized form of RNN that has fewer gates than the LSTM. Its model is very simple and clear, and it is becoming increasingly popular. GRU incorporates one gate, where the forget gate and input gate are combined into a single gate called the “update gate.” There is another “reset gate” added to the network structure, making the GRU easier to use by collaborating together [39]. Interestingly, GRU modulates data within the unit without using a separate memory cell. There are two activations at each time step: h_{t-1}^j and h_t^j . A linear interpolation is made between these two activations to figure out h_t^j . The update gate z_t^j determines which unit’s activation is updated based on this. Simultaneously, the candidate activation h_t^j is computed in order to update the gate. However, the update gate z controls the previous state value. The three sub-models will be all trained on safe (no faults) and benign (no attacks) states to generate an hybrid global framework. After that, they will be combined to perform the detection phase based on a thresholding function.

C. Detection: Thresholding and Error Mitigation

The thresholding operation is an acquired step that helps in the identification of abnormalities. In addition, there are available bases with sufficient training sets to help the thresholding detect abnormalities such as faults and/or attacks. However, our framework applies adaptive thresholding that takes into consideration the dynamic error threshold. Generally, the threshold process

aims to identify abnormal states by setting a thresholding extent ϵ . The threshold extent is accordingly established via a process to adapt with the current hyperparameters of the DL-model. It is an autonomous initiative that provides significant performance with minimal complexity. Thresholding does not require labeled data or empirical assumptions about error probabilities. The set extent of the threshold ϵ will quantify the detection of faults and attacks. The process is done via using the trained DL-model, where the thresholding can effectively identify abnormal states compared to safe and benign states. However, the detection performance varies along with the thresholding extent ϵ . When the detection error is unusually high or low, it appears important to support the framework so that it can be used to obtain an accurate detection. Therefore, the framework needs an extreme simplified form of the original abnormal detection problem. Based on this, dynamic error e and smoothing thresholds are a combined solution that stands for automatically identifying error values without relying on random assumptions. For each one-dimensional vector, errors and smoothing calculates the appended mistakes. In addition, this function handles and dampens the sharp spikes of the accidental errors that occur during the thresholding. After generating the output value $Y(t)$ for each timestamp t , the error e is calculated by $e(t) = |y^t - Y^t|$, where $y^t = x_i^{t+1}$ and i corresponds to the dimension of the true values. Alternatively, we add a recent function to mitigate fake alerts adopted from recent research [40] and combined with our framework. We propose a minimum score $s_{e\min}$ based on similar magnitude errors of s_e . These errors do not occur frequently within the same flow. This function aims to classify future errors as nominal when the set of s_e is less than or equal to $s_{e\min}$. The ground truth data is used to set the vector of $s_{e\min}$ for given networks and channels. This function helped to eliminate fake alerts that count rare UAVs behaviors at periodic intervals as failures or anomalies.

VI. SIMULATION STUDY AND RESULTS

In this section, we present the implementation of our framework detailed in Section V. We performed extensive experiments in order to validate the effectiveness of our framework.

A. Simulation Environment and Metrics

To accomplish the experiments, Python-TensorFlow [41] was employed to construct the DL-model. The full experiments leveraged an extensive number of libraries in the Python programming language. These codes run on a virtual machine (VM) environment with the following specifications: AMD Ryzen 5 CPU, AMD Radeon Vega 8 graphics with 12 GB of RAM. For the DL-models, we selected the best hyper-parameter values, which include the activation function, the optimizer, the error function, the loss function, the learning rate, which is set at 10^{-3} , and a dense activation layer. After we set all these training variables, we configure the training epochs to 900 epochs. We train each DL-sub-model with safe and benign states for two different cases of centralized and decentralized UAVs architectures. In the next step, we look at the general DL-model in different failure and attack states for both architectures. To validate our

TABLE II
EVALUATION OF ATTACKS DETECTION RESULTS USING ACCURACY METRICS

DL-Model	Dataset	Attack Type	Accuracy
LSTM	UA	Jamming attack	0.97
		Spoofing attack	0.97
		DoS	0.95
B-LSTM	CAIDA	Jamming attack	0.97
		Spoofing attack	0.98
		DoS	0.97
GRU	UA	Jamming attack	0.97
		Spoofing attack	0.97
		DoS	0.96

TABLE III
EVALUATION FOR FAULTS DETECTION RESULTS USING ACCURACY METRICS

Dataset	DL-Model	Detection Accuracy			
		Engine Failure	Elevator Failure	Ailerons Failure	Rudder Failure
ALFA	LSTM	0.83	0.83	0.85	0.87
	B-LSTM	0.90	0.89	0.91	0.93
	GRU	0.79	0.81	0.78	0.82

proposed model, we use different datasets composed of data samples under normal and attack scenarios. These datasets are public and available on IEEE DataPort such as AirLab Failure and Anomaly (ALFA) which includes different types of UAV failures and UAV attacks (UA) composed of jamming and spoofing attacks. We processed these datasets, and we obtained a list of selected features. To work on the challenges of the decentralized architecture, we attempted to approximate other datasets such as IoT network attacks including CAIDA dataset available in DataPort for research purposes. To make sure that our hybrid-AFFA framework is working properly, we pick them to cover a wide range of attacks. Different metrics are used to evaluate the performance of our AHFFA-UAVs. The following parameters are considered during the validation: true negative (TN), true positive (TP), false negative (FN), and false positive (FP). Based on these parameters, we consider the accuracy metrics as the most critical performance indicator for faults and attacks detection. It represents the proportion of the total number of correct failures and attacks detected for the UAV: roll, yaw, and pitch.

B. Simulation Results and Evaluation

The simulations demonstrate the efficient performance of our proposed framework that gives good accuracy value for spoofing, jamming and DoS attacks, and four types of failure. In Tables II and III, we present the detection results for UAVs attacks and faults in terms of accuracy with three sub-models, respectively. As we can observe in the attack's detection (see Fig. 6), the first sub-model-based LSTM outperforms well, where the accuracy of the general model reached 96.3% for attacks detection and 84.5% for faults detection. Thus, our choice of the LSTMs in the current experiments has been validated with good performance on jamming, spoofing, and DoS attacks. The accuracy results reached 97.3% and 90.7% for attacks and faults detection, respectively. This is because the recorded flight logs show that the sequences of abnormal behavior occur at different temporal behaviors than normal sequences. The B-LSTM sub-model works hard to detect different abnormal sequences

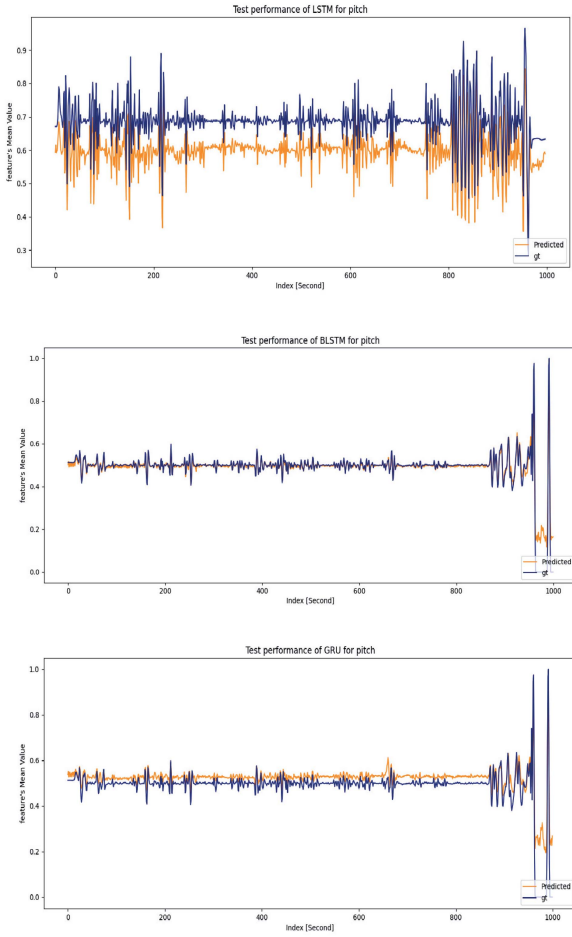


Fig. 4. Representative display of prediction detector based on UAVs pitch movement with error spikes of attacks sub-models LSTM, B-LSTM, and GRU.

that have similar temporal behavior. Through the experiments, we have understood that the difference between the LSTM and the GRU units is not so far for them to be similar. The GRU third sub-model reached 96.6% on attacks detection and 80% on faults detection. Thus, the results show that the LSTM is comparatively similar to the GRU, but still the LSTM is more accurate in faults detection. Compared to the basic LSTM, the GRU has fewer training parameters, runs faster, and uses less storage space. In order to evaluate our approach, we predict the pitch movement of the UAV under three attacks and four failures. The plot results of the attacks and the faults simulations for the test mission are shown in Figs. 4 and 5, respectively. The large spike towards the attack's test mission graph flags the flight as abnormal. We compare the prediction of the UAV test results to the ground truth data. These pitch translational motion measurements show a good predictive basis to announce the performance of the framework compared to the ground truth data at the exact time. Based on this, this approach can ensure a good prediction for both attacks and faults detection. In Fig. 4 the predictor is depicted as a line that represents the LSTM, B-LSTM, and GRU performance on attacks detection compared to the ground truth. In the first sub-model plot, it is seen that the prediction is good in the LSTM model. The second sub-model plot shows a better prediction for B-LSTM. There is a slight low

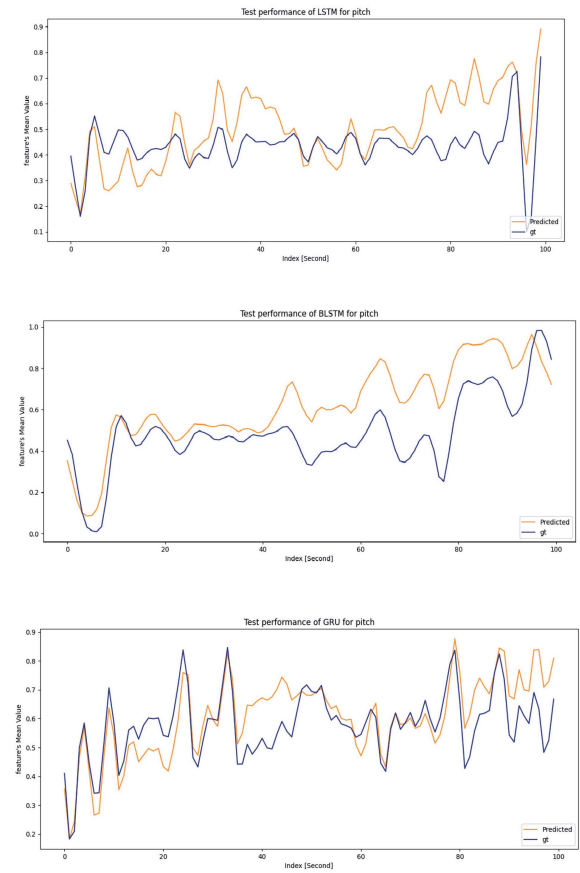


Fig. 5. Representative display of prediction detector based on UAVs pitch movement with error spikes of faults sub-models LSTM, B-LSTM, and GRU.

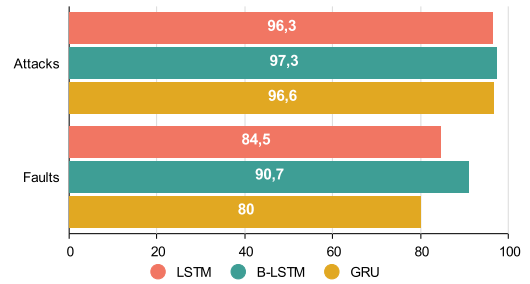


Fig. 6. BoxPlots of accuracy results for faults and attacks detection using LSTM, B-LSTM, and GRU DL-models.

performance for the third sub-model plot referred to as the GRU. We can also see that the prediction converges with the ground truth data with the number of epochs. This shows that the higher accuracy in B-LSTM models is not because of a greater number of epochs, but due to better model architecture. In addition, Fig. 5 shows the reconstruction of the prediction validation by the three sub-models. It can be seen that the detection performance of faults is desirable. It is to be noted that the accuracy reached 84.5%, 90.7%, and 80%, for the LSTM, B-LSTM and GRU models, respectively. In addition, it is observed that when trying to evaluate sequences with failure, the B-LSTM model mostly gives out the best performance. This shows that the B-LSTM sub-model has learned more information in bidirectional directions. We discuss how the threshold for detecting UAV faults and

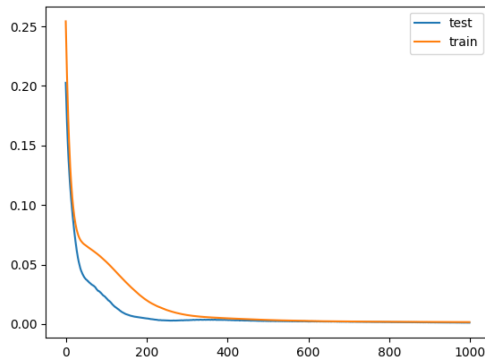


Fig. 7. AHFA-UAVs framework train and validation learning loss curves.

attacks is determined based on the distribution of reconstruction losses for validation. The best performance of our hybrid-AFFA framework is observed in Fig. 7 that shows the loss distribution for our hybrid-AFFA framework. We observe through the graph that the loss converges after a few epochs for both faults and attacks detection. The result curves for loss emphasize the performance of our framework with a low loss.

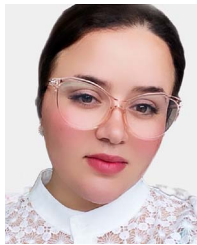
VII. CONCLUSION AND FUTURE WORK

In this article, we proposed a secure, efficient, and intelligent framework named adaptive hybrid UAVs framework for faults and attacks. The basis technique of our framework is built using three different alternatives to DL models. We trained these sub-models and combined their power using simulation datasets to demonstrate the detection performance for an adaptive environment that includes centralized and decentralized architectures. We summarize the performance for the three different sub-models where they show that, unlike expected, LSTM outperforms well and the B-LSTM can learn information and achieve better performance. After simulations and evaluations, empirical results indicate that our framework has a detection rate of over 85% and 96,7% for UAVs faults and attacks, respectively.

REFERENCES

- [1] F. Tlili, L. C. Fourati, S. Ayed, and B. Ouni, "Investigation on vulnerabilities, threats and attacks prohibiting UAVs charging and depleting UAVs batteries: Assessments & countermeasures," *Ad Hoc Netw.*, vol. 129, 2022, Art. no. 102805.
- [2] H. Shakhateh et al., "Unmanned aerial vehicles (UAVs): A survey on civil applications and key research challenges," *IEEE Access*, vol. 7, pp. 48572–48634, 2019.
- [3] S. Zaidi, M. Atiquzzaman, and C. T. Calafate, "Internet of flying Things (IoFT): A survey," *Comput. Commun.*, vol. 165, pp. 53–74, 2021.
- [4] G. Loukas, T. Vuong, R. Heartfield, G. Sakellari, Y. Yoon, and D. Gan, "Cloud-based cyber-physical intrusion detection for vehicles using deep learning," *IEEE Access*, vol. 6, pp. 3491–3508, 2017.
- [5] A. Haddaji, S. Ayed, and L. C. Fourati, "Artificial intelligence techniques to mitigate cyber-attacks within vehicular networks: Survey," *Comput. Elect. Eng.*, vol. 104, 2022, Art. no. 108460.
- [6] G. Choudhary, V. Sharma, I. You, K. Yim, R. Chen, and J.-H. Cho, "Intrusion detection systems for networked unmanned aerial vehicles: A survey," in *Proc. 14th Int. Wireless Commun. Mobile Comput. Conf.*, 2018, pp. 560–565.
- [7] Y. Mekdad et al., "A survey on security and privacy issues of UAVs," 2021, *arXiv:2109.14442*.
- [8] A. Saif et al., "An efficient energy harvesting and optimal clustering technique for sustainable postdisaster emergency communication systems," *IEEE Access*, vol. 9, pp. 78188–78202, 2021.
- [9] O. Shrit, S. Martin, K. Alagha, and G. Pujolle, "A new approach to realize drone swarm using ad-hoc network," in *Proc. 16th Annu. Mediterranean Ad Hoc Netw. Workshop*, 2017, pp. 1–5.
- [10] M. Bekhti, M. Garraffa, N. Achir, K. Boussetta, and L. Létocart, "Assessment of multi-UAVs tracking for data gathering," in *Proc. 13th Int. Wireless Commun. Mobile Comput. Conf.*, 2017, pp. 1004–1009.
- [11] S. Jeong, E. Park, K. U. Seo, J. D. Yoo, and H. K. Kim, "MUVIDS: False MAVLink injection attack detection in communication for unmanned vehicles," in *Proc. Workshop Automot. Auton. Veh. Secur.*, 2021, pp. 25.
- [12] A. Allouch, O. Cheikhrouhou, A. Koubâa, M. Khalgui, and T. Abbas, "MAVSec: Securing the MAVLink protocol for Ardupilot/PX4 unmanned aerial systems," in *Proc. 15th Int. Wireless Commun. Mobile Comput. Conf.*, 2019, pp. 621–628.
- [13] I. U. Khan, S. B. Hussain Shah, L. Wang, M. A. Aziz, T. Stephan, and N. Kumar, "Routing protocols & unmanned aerial vehicles autonomous localization in flying networks," *Int. J. Commun. Syst.*, 2021, Art. no. e4885.
- [14] A. Allouch, O. Cheikhrouhou, A. Koubâa, M. Khalgui, and T. Abbas, "MAVSec: Securing the MAVLink protocol for Ardupilot/PX4 unmanned aerial systems," in *Proc. 15th Int. Wireless Commun. Mobile Comput. Conf.*, 2019, pp. 621–628.
- [15] J. Qiu, D. Grace, G. Ding, M. D. Zakaria, and Q. Wu, "Air-ground heterogeneous networks for 5G and beyond via integrating high and low altitude platforms," *IEEE Wireless Commun.*, vol. 26, no. 6, pp. 140–148, Dec. 2019.
- [16] J. Qiu, D. Grace, G. Ding, J. Yao, and Q. Wu, "Blockchain-based secure spectrum trading for unmanned-aerial-vehicle-assisted cellular networks: An operator's perspective," *IEEE Internet Things J.*, vol. 7, no. 1, pp. 451–466, Jan. 2020.
- [17] R. Zhang, J.-P. Condomines, and E. Lochin, "A multifractal analysis and machine learning based intrusion detection system with an application in a UAS/RADAR system," *Drones*, vol. 6, no. 1, pp. 21, 2022.
- [18] Q. A. Al-Haija and A. A. Badawi, "High-performance intrusion detection system for networked UAVs via deep learning," *Neural Comput. Appl.*, vol. 34, no. 13, pp. 10885–10900, 2022.
- [19] J. Whelan, A. Almhadi, and K. El-Khatib, "Artificial intelligence for intrusion detection systems in unmanned aerial vehicles," *Comput. Elect. Eng.*, vol. 99, 2022, Art. no. 107784.
- [20] A. Wang, W. Wang, H. Zhou, and J. Zhang, "Network intrusion detection algorithm combined with group convolution network and snapshot ensemble," *Symmetry*, vol. 13, no. 10, pp. 1814, 2021.
- [21] R. A. Ramadan, A.-H. Emar, M. Al-Sarem, and M. Elhamamy, "Internet of drones intrusion detection using deep learning," *Electronics*, vol. 10, no. 21, pp. 2633, 2021.
- [22] B. Chen, Y. Peng, B. Gu, Y. Luo, and D. Liu, "A fault detection method based on enhanced GRU," in *Proc. Int. Conf. Sens. Meas. Data Analytics Era Artif. Intell.*, 2021, pp. 1–4.
- [23] K. H. Park, E. Park, and H. K. Kim, "Unsupervised fault detection on unmanned aerial vehicles: Encoding and thresholding approach," *Sensors*, vol. 21, no. 6, pp. 2208, 2021.
- [24] C. Titouna, F. Naït-Abdesselam, and H. Moungra, "An online anomaly detection approach for unmanned aerial vehicles," in *Proc. Int. Wireless Commun. Mobile Comput.*, 2020, pp. 469–474.
- [25] J. Whelan, T. Sangarapillai, O. Minawi, A. Almhadi, and K. El-Khatib, "Novelty-based intrusion detection of sensor attacks on unmanned aerial vehicles," in *Proc. 16th ACM Symp. QoS Secur. Wireless Mobile Netw.*, 2020, pp. 23–28.
- [26] K. Xiao, J. Zhao, Y. He, C. Li, and W. Cheng, "Abnormal behavior detection scheme of UAV using recurrent neural networks," *IEEE Access*, vol. 7, pp. 110293–110305, 2019.
- [27] M. P. Arthur, "Detecting signal spoofing and jamming attacks in UAV networks using a lightweight IDS," in *Proc. Int. Conf. Comput. Inf. Telecommun. Syst.*, 2019, pp. 1–5.
- [28] B. Lindemann, F. Fesenmayr, N. Jazdi, and M. Weyrich, "Anomaly detection in discrete manufacturing using self-learning approaches," *Procedia CIRP*, vol. 79, pp. 313–318, 2019.
- [29] D. Muniraj and M. Farhood, "A framework for detection of sensor attacks on small unmanned aircraft systems," in *Proc. Int. Conf. Unmanned Aircr. Syst.*, 2017, pp. 1189–1198.
- [30] S. M. Kasongo and Y. Sun, "A deep long short-term memory based classifier for wireless intrusion detection system," *ICT Exp.*, vol. 6, no. 2, pp. 98–103, 2020.
- [31] K. M. Siddiqui and S. Naahid, "Analysis of KDD CUP 99 dataset using clustering based data mining," *Int. J. Database Theory Appl.*, vol. 6, no. 5, pp. 23–34, 2013.
- [32] D. D. Protić, "Review of KDD CUP 99, NSL-KDD and kyoto 2006 datasets," *Vojnotehnički glasnik/Military Tech. Courier*, vol. 66, no. 3, pp. 580–596, 2018.

- [33] L. Dhanabal and S. P. Shanharajah, "A study on NSL-KDD dataset for intrusion detection system based on classification algorithms," *Int. J. Adv. Res. Comput. Commun. Eng.*, vol. 4, no. 6, pp. 446–452, 2015.
- [34] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in *Proc. Mil. Commun. Inf. Syst. Conf.*, 2015, pp. 1–6.
- [35] R. Panigrahi and S. Borah, "A detailed analysis of CICIDS2017 dataset for designing intrusion detection systems," *Int. J. Eng. Technol.*, vol. 7, no. 3.24, pp. 479–482, 2018.
- [36] A. R. Gad, A. A. Nashat, and T. M. Barkat, "Intrusion detection system using machine learning for vehicular ad hoc networks based on ToN-IoT dataset," *IEEE Access*, vol. 9, pp. 142206–142217, 2021.
- [37] A. Haddaji, S. Ayed, and L. Chaari, "Federated learning with blockchain approach for trust management in IoV," in *Proc. Int. Conf. Adv. Inf. Netw. Appl.*, Springer, 2022, pp. 411–423.
- [38] M. Chhetri, S. Kumar, P. P. Roy, and B.-G. Kim, "Deep BLSTM-GRU model for monthly rainfall prediction: A case study of Simtokha, Bhutan," *Remote Sens.*, vol. 12, no. 19, pp. 3174, 2020.
- [39] E. Ahmadzadeh, H. Kim, O. Jeong, N. Kim, and I. Moon, "A deep bidirectional LSTM-GRU network model for automated ciphertext classification," *IEEE Access*, vol. 10, pp. 3228–3237, 2022.
- [40] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using LSTMs and nonparametric dynamic thresholding," in *Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2018, pp. 387–395.
- [41] D. Rothman, *Transformers for Natural Language Processing: Build Innovative Deep Neural Network Architectures for NLP With Python, PyTorch, TensorFlow, BERT, RoBERTa, and More*. Birmingham, U.K.: Packt Publishing Ltd, 2021.



Fadhila Tlili received the master's degree in computer science and multimedia from the Computer Science and Multimedia Higher Institute of Sfax (ISIMS), Tunisia, in 2020. She is currently working toward the PhD degree with Computer Science and Communication Sciences and Technologies, National School of Electronics and Telecommunications of Sfax (ENET'Com). She is a member of the Laboratory of Signals, systems, artificial Intelligence, neTworkS (SM@RTS), Sfax Digital Research Center.

Her scope of research ranges from security mechanisms, which are specially related to autonomous systems comprising UAVs, drones systems and Artificial intelligence merged techniques. The primary objective of her PhD research project is to design and validate an intelligent and secure UAV framework using new security approaches to manage security requirements in UAV context. She works on different intelligent approaches based on artificial intelligence techniques (machine learning, deep learning, game theory, etc.) that should be considered to optimize the QoE and QoS of the different proposals.



Samiha Ayed received the BS and MS degrees in computer science from the National School of Computer Science of Manouba – Tunisia, respectively, in 2005 and 2006, and the PhD degree in security of information systems from Telecom Bretagne, Rennes, France, in 2009. From 2012 to 2016, he was a researcher with the SFIIS team of Telecom Bretagne. Since 2017, she has been an associate professor with the HETIC Department, University of Technology of Troyes. Her research activities concern many axes about cybersecurity challenges (privacy, trust management, access control) using the emerging paradigms like the blockchain technology, the artificial intelligence and the fog/cloud computing. The application domains of her research activities are mainly the IoT; the IoV; the vehicular environments as well as the e-health environment.



Lamia Chaari Fourati is a professor with Computer Science and Multimedia Higher Institute and Researcher Laboratory of Technologies for Smart Systems (LT2S), Sfax Digital Research Center. She focused her research activities on conception and validation of new protocols and mechanisms for emerging networks technologies. Her research activities are very important and up-to-date which are related to digital telecommunication networks, in particular wireless access networks, sensor networks, vehicular networks, Internet of Things, 5G, software defined

network, information centric network and wireless body area network, unmanned aerial vehicle. In these areas, she is interested in problems such as quality of services provisioning (congestion control, admission control, resources allocations.), cybersecurity and ambient intelligence. Furthermore, she focused her research on applications that impact positively our society such as healthcare domain, through the conception of innovative protocols and mechanisms for health monitoring, remote systems, the environmental control as well as energy saving approaches. Her scientific publications have met the interest of the scientific community and her work has been published in a very good journal and conferences. She has more than 60 papers published in journals and more than 90 papers published in conferences. She is the laureate for the Kwame Nkrumah Regional Awards for women 2016 (North Africa Region).